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DSC 140S

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NBA Playoffs 2023-2024 Report

Introduction

The National Basketball Association is the official professional basketball league, where data plays a crucial role in player evaluation and team decision-making. The dataset used for this project includes detailed statistics for NBA players across the 2023-2024 playoff season (Shouqi). Here are all the shortened column names that will be used in this report:

- DRtg (defensive rating): An estimate of how many points a player allows the other team to score while they are playing on defense.
- POS (position): The specific role a player plays on the team, which affects where and how they play on the court.
- 3PA (3 pointers attempted): The number of times a player tries to score by shooting from the three-point line.
- TS% (true shooting %): A measure of how efficiently a player scores, combining all types of shots into one accuracy number. For example, if a player had to shoot a shot because time was running out, it might not count as much as if someone was wide open.
- PPG (points per game): The average number of points a player scores in each game.
- RPG (rebounds per game): The average number of times per game a player retrieves the ball after a missed shot.
- APG (assists per game): The average number of times per game a player helps a teammate score by passing them the ball.

This analysis aims to uncover insights from three questions: Does age affect a player's defensive rating differently across positions? Is there a relationship between three-points attempted and true shooting percentage? Finally, can a player's position be predicted based on their scoring, rebounding, and assists per game? The first question helps to reveal whether aging impacts defensive effectiveness unequally across positions, which can inform team decisions on player development and rotation strategies. As the NBA emphasizes three-point shooting, the second question is important to know when players start to get their groove with shooting three pointers. This insight can guide offensive strategies and evaluate whether high-volume shooters are contributing effectively. The last question explores whether traditional box score stats still align with positional roles in today's evolving game. A predictive relationship would support the use of core stats in player classification and scouting. Overall, this analysis will help uncover how player performance metrics relate to age, efficiency, and position—providing valuable insights for the modern NBA.

Methods

This analysis was conducted using Jupyter Notebook, which was the program used to run Python code along with data visualizations. The following Python libraries and functions were used in the project:

- Pandas was used to import and manipulate the dataset using DataFrames.
- Matplotlib.pyplot was used to create nicer visual graphs and plots to help present the data.
- Scipy.stats included functions like pearsonr, spearmanr, f_oneway, and ttest_ind, and were used to calculate correlations and perform statistical tests.
- Scikit-learn was used for the machine learning tasks and included functions like KNeighborsClassifier, train_test_split, accuracy_score, confusion_matrix, ConfusionMatrixDisplay, and RandomForestClassifier to perform the tests.

Data

To begin the analysis, the dataset was loaded using read_csv and cleaned by dropping any null values using dropna(). Basic information about the dataset was checked using info(). Looking at figure 1, every column had 213 rows filled with information, so the dataset is now ready to perform analysis on the research questions.

FIGURE 1

```
Data columns (total 29 columns):
 #   Column  Non-Null Count  Dtype  
---  ------  --------------  -----  
 0   RANK    213 non-null    int64  
 1   NAME    213 non-null    object 
 2   TEAM    213 non-null    object 
 3   POS     213 non-null    object 
 4   AGE     213 non-null    float64
 5   GP      213 non-null    int64  
 6   MPG     213 non-null    float64
 7   USG%    213 non-null    float64
 8   TO%     213 non-null    float64
 9   FTA     213 non-null    int64  
 10  FT%     213 non-null    float64
 11  2PA     213 non-null    int64  
 12  2P%     213 non-null    float64
 13  3PA     213 non-null    int64  
 14  3P%     213 non-null    float64
 15  eFG%    213 non-null    float64
 16  TS%     213 non-null    float64
 17  PPG     213 non-null    float64
 18  RPG     213 non-null    float64
 19  APG     213 non-null    float64
 20  SPG     213 non-null    float64
 21  BPG     213 non-null    float64
 22  TPG     213 non-null    float64
 23  P+R     213 non-null    float64
 24  P+A     213 non-null    float64
 25  P+R+A   213 non-null    float64
 26  VI      213 non-null    float64
 27  ORtg    213 non-null    float64
 28  DRtg    213 non-null    float64
dtypes: float64(21), int64(5), object(3)
```

For the first research question, the goal was to determine if the age of a player affects the player's defensive rating across the three main positions: Guard, Forward, and Center. This dataset also included secondary positions, but since the focus was on the main positions, a function was created to figure out what everyone's primary positions were, and group them by only their primary position. The new column that held this information was called pos_category, which indicates the primary position category.

Next, the average, median, and standard deviation for age and defensive rating were calculated within each primary position group using the groupby() command. Looking at figure 2, the average age for each position is in the ballpark of 27 years old and the average defensive rating for each position isn't all the same. The forward position seems to be a lot lower than the other two positions, but that is because a lot of players had 0, since they did not have a lot of playing time to get a defensive rating.

FIGURE 2

Age Stats for each Position:

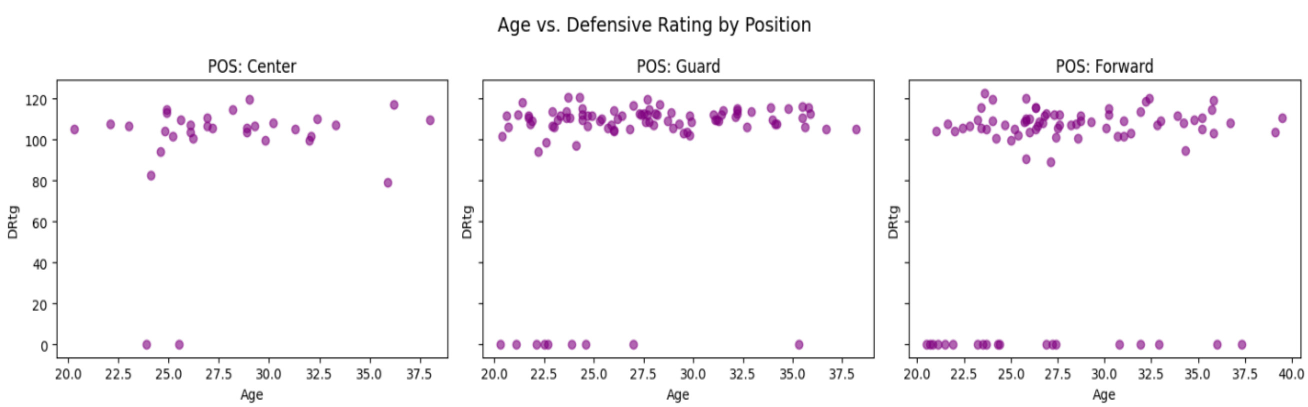
POS_category	mean	median	std
Center	27.993939	26.9	4.178587
Forward	27.957471	26.9	4.733813
Guard	27.444086	27.0	4.584673

Defensive Rating Stats for each Position:

POS_category	mean	median	std
Center	98.748485	105.6	26.738550
Forward	84.440230	106.3	45.237902
Guard	99.351613	109.6	33.017277

To get a visualization of this data to get a better understanding, I used a for loop to create three scatterplots, one for each position. Looking at figure 3, with forward now being visualized, it shows why it was a lot lower, as they had more people at 0 defensive rating. Studying these plots closer, all three seem to have a slope of close to 0 but can be looked further into by finding Pearson and Spearman correlation coefficients.

FIGURE 3



Pearson and Spearman correlations help to identify if two columns have some sort of connection (Spearmanr#). After calculating the coefficients, figure 4 shows that every correlation is between .19-.23, further helping that there is a slight positive correlation, but not much.

FIGURE 4

```
Correlation Results by Position:
Center: Pearson = 0.19, Spearman = 0.19
Guard: Pearson = 0.23, Spearman = 0.22
Forward: Pearson = 0.21, Spearman = 0.23
```

To finally conclude the problem, a one-way ANOVA test was conducted to figure out if there was any significance between the two columns. This test compares the means of groups to determine if one of the groups is significantly different from the others (Hamelg). After conducting the test, the p-value is .7109, which is greater than .05, concluding that age and defensive rating do not have a significant relationship.

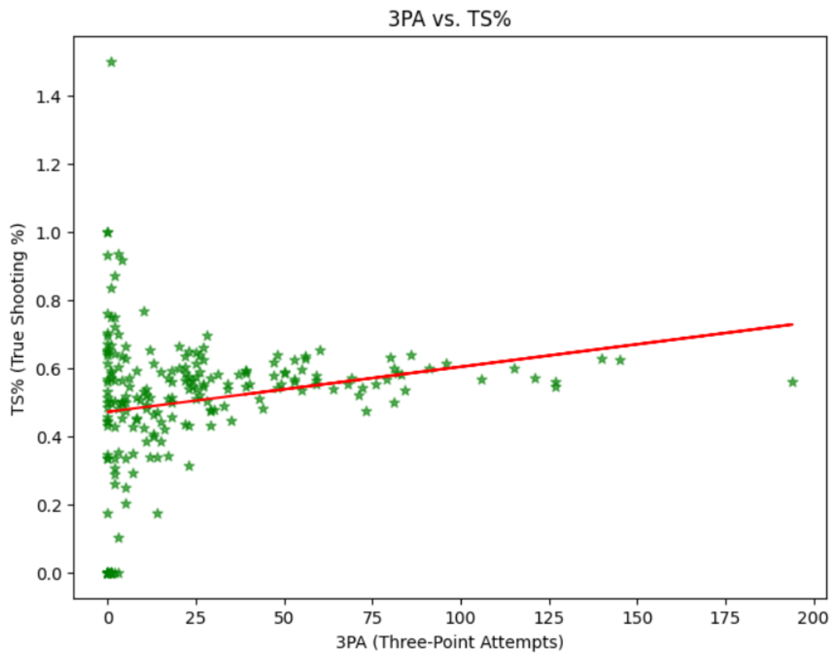
Moving into the second question, is there a relationship between three-point attempts and true shooting percentage? To start off, the mean, median, and standard deviation (std) for both columns was found. Figure 5 visualizes the information found and gives great insight, especially looking at the means of both. Understanding the mean true shooting percentage can help prove when players start to find their groove. Also, the average three-point attempts is 13, which is a fair number of attempts to further investigate.

FIGURE 5

```
3PA Stats:
 mean      24.985915
median    13.000000
std       32.278633
Name: 3PA, dtype: float64
True Shooting % Stats:
 mean      0.504197
median    0.544000
std       0.207132
Name: TS%, dtype: float64
```

Next, a linear regression line was made to further put onto a scatter plot. Figure 6 shows a scatter plot of three points attempts versus true shooting percentage, which visualizes how many shots it takes for a player to find their groove. Once a player starts to find their groove is when a player is attempting more shots, but still staying around the mean true shooting percentage.

FIGURE 6

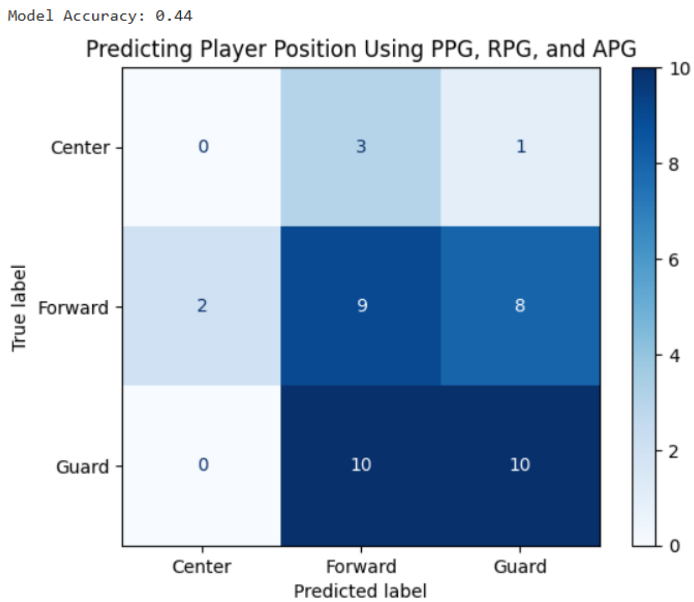


To further find out how related these statistics are, Pearson's and Spearman's correlation coefficients were calculated, and ended up with .206 for the Pearson coefficient and .231 for the Spearman coefficient. This can show a slight positive correlation, but looking at our scatter plot above, that could have been heavily skewed by the number of people that did not make a three-point attempt.

To conclude this problem, a two-sample t-test was conducted using high vs. low three points attempt clusters. To start, the median value was found, and the low and high splits were created from the median value, setting the low cluster less than the median and the high cluster to be greater. After conducting the test, the p-value was .0018. That shows that there is a significant relationship between true shooting % and three point-attempts.

Finally, can a player's position be predicted based on their scoring, rebounding, and assists? To solve this problem, a random forest classifier test was conducted. This test works by creating multiple decision trees using different groups of data, then combining their predictions to make a final classification (Shafi). To do that, the data was split into two groups: a test group and a train group. First, we put the train group into the random forest classifier and then ran the test group to see how well the train group did. Looking at figure 7, a confusion matrix is made to visualize how the test group did compared to the train group. The accuracy was 44% which means the test correctly categorized 44% of the correct positions based on the player's points, rebounds, and assists per game. The confusion matrix shows the grid of true vs predicted guesses, and if the position groups line up, those are the correct answers. For example, the predicted guessed Center for 2 players, but they were forwards.

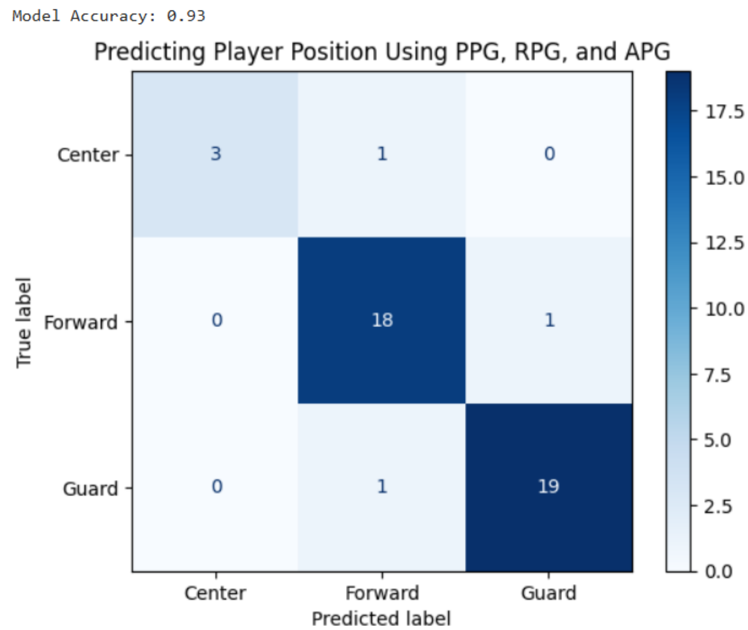
FIGURE 7



To improve model accuracy, cross-validation and manual grid search were used. First, a hyperparameter grid was created to figure out the best value for n_estimators and max_depth. N_estimators controls the number of decision trees and max_depth limits how deep each tree can grow (Shafi). Then, a for loop was created to test each combination of estimators and depth. For cross-validation, 5 random splits were used, and figure 8 shows the results. After looking at the results, it seems that estimators 100 and depth 10 were the best parameters for the test.

After re-running the test, figure 8 shows that the model accuracy is now 93%. That is a very good accuracy score, and the confusion matrix also helps in expressing the accuracy. To conclude, 93% is a very accurate score and the random forest classifier test was a good one to choose for this problem.

FIGURE 8



Results

Starting off with the first question, the analysis showed no significant relationship between a player's age and their defensive rating within the guard, forward, or center positions. Although a slight positive correlation was observed, the one-way ANOVA test confirmed that the variation was not statistically significant. This suggests that age by itself is not a strong predictor of defensive performance, which is interesting because it challenges the common belief that younger players are automatically better defenders.

For the second question, a weak but statistically significant positive correlation was found between three-point attempts and true shooting percentage. The scatterplot indicated that players who took more three-point shots often maintained similar or better shooting efficiency. The t-test between high and low attempt groups confirmed a meaningful difference. This is valuable because it shows that high-volume shooters aren't necessarily less efficient—many may be experienced or skilled enough to maintain accuracy even with a high shooting load.

Finally, an algorithm was made to find out if a player's position can be predicted based on points, rebounds, and assists per game. Initially, the random forest model predicted positions with only 44% accuracy, but after tuning the hyperparameters with cross-validation, the accuracy improved significantly to 93%. This strong result indicates that basic performance stats like PPG, RPG, and APG can be powerful predictors of a player's position when modeled correctly. It shows how machine learning can be applied in sports analytics for role identification or scouting purposes.

Conclusion

This analysis used playoff data from the 2023–2024 NBA season to explore how age, shooting behavior, and basic performance metrics relate to player effectiveness and positional roles. These insights are especially meaningful because playoff statistics reflect players' performance under high-stakes conditions, offering a clearer picture of who delivers when it matters most. The findings challenge assumptions, such as age being a reliable predictor of defense, and highlight how three-point volume doesn't necessarily reduce efficiency. Most notably, the ability to predict player position with 93% accuracy using just scoring, rebounding, and assist averages shows the practical power of machine learning in modern sports analytics. These results can support smarter scouting, lineup construction, and player development strategies, demonstrating the real-world value of data-driven decision-making in the NBA.

Works Cited

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